QUBO Formulation: Financial Fraud Feature Selection

Financial Fraud Feature Selection

Problem Statement. Given a set of N candidate features for fraud detection, each feature $i \in \{0, \ldots, N-1\}$ is associated with a predictive value $v_i \geq 0$. Each unordered pair of distinct features (i, j) with $i < j$ carries a redundancy penalty $r_{ij} \geq 0$. The task is to select exactly K features such that the total predictive value is maximized while subtracting the pairwise redundancy penalties among the selected features.

Decision Variables. For each candidate feature $i \in \{0, \ldots, N-1\}$, introduce a binary decision variable $x_i \in \{0, 1\}$, where $x_i = 1$ indicates that feature i is included in the selected set, and $x_i = 0$ otherwise.

QUBO Derivation. Step 1. The original problem is a maximization. To cast it as a QUBO minimization, we negate the objective function:

$$\min_{x \in \{0,1\}^N} \left(-\sum_{i=0}^{N-1} v_i x_i + \sum_{0 \leq i < j \leq N-1} r_{ij} x_i x_j \right). \tag{1}$$

Step 2. The cardinality constraint requires exactly K features to be selected:

$$\sum_{i=0}^{N-1} x_i = K. \tag{2}$$

Step 3. We enforce the constraint via a quadratic penalty term with weight $\lambda > 0$.

$$\text{(a) Squared penalty:} \quad \lambda \left(\sum_{i=0}^{N-1} x_i - K \right)^2 . \tag{3}$$

$$\text{(b) Expansion:} \quad \lambda \left(\sum_{i=0}^{N-1} x_i^2 - 2K \sum_{i=0}^{N-1} x_i + K^2 + \sum_{i \neq j} x_i x_j \right). \tag{4}$$

Using the idempotent property $x_i^2 = x_i$ and noting $\sum_{i \neq j} x_i x_j = 2 \sum_{0 \leq i < j \leq N-1}$

(5)

$$\lambda \left(\sum_{i=0}^{N-1} (1 - 2K) x_i + 2 \sum_{0 \leq i < j \leq N-1} x_i x_j + K^2 \right). \tag{6}$$

$$\text{(c) Penalty contribution to } H: \quad \lambda \sum_{i=0}^{N-1} (1 - 2K) x_i + 2\lambda \sum_{0 \leq i < j \leq N-1} x_i x_j + \lambda K^2. \tag{7}$$

Step 4. Combining the negated objective from Step 1 with the penalty contribution from Step 3 yields the full QUBO Hamiltonian $H(x)$:

$$H(x) = \sum_{i=0}^{N-1} \left(-v_i + \lambda(1 - 2K) \right) x_i + \sum_{0 \leq i < j \leq N-1} (r_{ij} + 2\lambda)\, x_i x_j + \lambda K^2. \tag{8}$$

Step 5. Reading off the coefficients from the general form $H(x) = \sum_i Q_{i,i} x_i + \sum_{i<j} Q_{i,j} x_i x_j + c$, we obtain the closed-form expressions for the upper-triangular Q matrix entries and the constant offset:

$$Q_{i,i} = -v_i + \lambda(1 - 2K), \quad \forall i \in \{0, \ldots, N-1\}, \tag{9}$$

$$Q_{i,j} = r_{ij} + 2\lambda, \quad \forall 0 \leq i < j \leq N-1, \tag{10}$$

$$c = \lambda K^2. \tag{11}$$

Penalty Weight Justification. Let $V_{\max} = \max_i v_i$ denote the maximum predictive value of any single feature. The maximum possible objective gain achievable by selecting any subset of features is bounded by $NV_{\max}$. To guarantee that constraint violation is never energetically favorable, the penalty weight λ must exceed the maximum possible improvement obtainable by an infeasible assignment. A sufficient condition is $\lambda > NV_{\max}$. In practice, choosing λ significantly larger than any individual v_i ensures that the energy cost of deviating from the cardinality constraint strictly dominates any potential gain from feature selection, thereby enforcing feasibility.

Correctness Argument. Minimizing $H(x)$ over $\{0,1\}^N$ recovers the optimal solution because (i) for any feasible assignment satisfying $\sum_{i=0}^{N-1} x_i = K$, the penalty term vanishes

and $H(x)$ reduces exactly to the negated original objective; thus, the global minimum over the feasible subspace corresponds to the maximum of the original objective. (ii) For any infeasible assignment, the penalty term contributes a strictly positive energy $\lambda(\sum_{i=0}^{N-1} x_i - K)^2 \geq \lambda$. By selecting λ sufficiently large, the energy of any infeasible point exceeds the energy of the optimal feasible point. Consequently, the global minimum of $H(x)$ must lie within the feasible region and correspond to the optimal feature set.

Illustrative Example. Consider a concrete instance with $N = 3$ candidate features and $K = 2$ required selections. Let the predictive values be $v_0 = 5, v_1 = 3, v_2 = 4$ and the redundancy penalties be $r_{01} = 1, r_{02} = 2, r_{12} = 1$. Using $\lambda = 10000$, the general formulas instantiate as follows.

The concrete objective terms are $-5x_0 - 3x_1 - 4x_2 + 1x_0x_1 + 2x_0x_2 + 1x_1x_2$. The concrete penalty terms expand to $10000\,((1-4)(x_0 + x_1 + x_2) + 2(x_0x_1 + x_0x_2 + x_1x_2) + 4)$. Combining these yields the explicit Hamiltonian:

$$\begin{aligned} H(x) &= (-5 - 20000)x_0 + (-3 - 20000)x_1 + (-4 - 20000)x_2 \\ &\quad + (1 + 20000)x_0x_1 + (2 + 20000)x_0x_2 + (1 + 20000)x_1x_2 + 40000 \\ &= -20005x_0 - 20003x_1 - 20004x_2 + 20001x_0x_1 + 20002x_0x_2 + 20001x_1x_2 + 40000. \end{aligned} \tag{12}$$

The corresponding Q matrix entries are $Q_{0,0} = -20005$, $Q_{1,1} = -20003$, $Q_{2,2} = -20004$, $Q_{0,1} = 20001$, $Q_{0,2} = 20002$, $Q_{1,2} = 20001$, with offset $c = 40000$.

Evaluating all $\binom{3}{2} = 3$ feasible assignments:

- $x = (1, 1, 0)$: $H = -20005 - 20003 + 20001 + 40000 = 39993$.
- $x = (1, 0, 1)$: $H = -20005 - 20004 + 20002 + 40000 = 39993$.
- $x = (0, 1, 1)$: $H = -20003 - 20004 + 20001 + 40000 = 39994$.

The minimum energy is 39993, achieved by $x = (1, 1, 0)$ and $x = (1, 0, 1)$. These correspond to feature sets $\{0, 1\}$ and $\{0, 2\}$, both yielding a maximum total score of 7.